CHANDU PRASAD BHAIRAPU



AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Neu-Isenburg • EU Blue Card
linkedin github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in data-driven solutions from problem definition to evaluation, combining machine learning and forecasting with generative AI and RAG. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by data science work on industrial process data.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Owned a knowledge-extraction solution end-to-end as sole developer, from data preparation and chunking to prompt engineering and indexing, reporting directly to the CTO and translating the needs of SAP domain experts into the technical approach.
- Built multi-agent extraction with LangGraph and evaluated it systematically against an SAP-system-derived ground truth, raising precision/recall from 33%/45% to 93%/90%.
- Chose a deterministic, code-based pipeline over LLM calls for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Built an MCP service exposing the knowledge graph for millisecond-latency retrieval, and Expert Interface, a web app (React, Python microservices) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, Neo4j, MongoDB, Postgres, FastAPI, Docker, Kubernetes, Azure, React**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised where agentic AI could support them and translated their requirements into the solution approach.
- Designed and deployed an agentic AI application with LangGraph (vector database queries, general inference, summarization) with a FastAPI backend and React frontend; Docker, AWS CDK and GitLab CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Designed the evaluation of the PrismaChat RAG system: automated generation and retrieval metrics with RAGAS and PytrecEval against chemist-provided ground truth, on AWS infrastructure (Lambda, SQS, DynamoDB) built with CDK.
- Benchmarked multilingual embedding models on over 1M chunks of internal chemical data to choose the best method, and designed RAG pipelines (chunking, vector databases, indexing, fine-tuning models for retrieval).
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Scikit-Learn, AWS, Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (viscosity, reaction progression) from time-series process data; benchmarked against empirical curve-fitting, the models improved early identification of batches likely to deviate from spec.
- Engineered lag features and rolling statistical windows from sensor time series for the tree models and fed raw sequences into the LSTM; data preparation and analysis with Pandas, NumPy and SciPy.
- Delivered a Streamlit dashboard where chemists simulate parameter changes and see predictions before running physical batches, turning model results into recommendations they could act on.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM forecasting models for EV load time series; the LSTM reached 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Analysis of Life Expectancy with Regression Models | Python | Scikit-Learn

- Statistical analysis of life expectancy data with regression models, visualization and feature selection to identify the main influencing factors.

Project: Clinical Info Assistant | Python | FastAPI | LLMs | React

- Web application that answers questions about medicines from drug label texts, with separate user journeys for healthcare professionals and patients, cited sources on every answer, and a handoff when a patient question needs a doctor.
- Python FastAPI backend with retrieval and a grounding check; usage events and a funnel view to study user behavior; labelled evaluation set run in CI.

TECHNICAL SKILLSET

- **Data Science & Statistics:** Python, SQL, Pandas, NumPy, SciPy, Matplotlib, feature engineering, regression, model evaluation, time-series forecasting (ARIMA, SARIMA, Prophet, LSTM)
- **Machine Learning:** Scikit-Learn, XGBoost, Random Forest, PyTorch, TensorFlow/Keras, Hugging Face Transformers
- **Generative AI:** Large Language Models, RAG, GraphRAG, AI agents (LangGraph, LangChain), prompt engineering, LiteLLM, MCP, RAGAS, QLoRA fine-tuning
- **Engineering & Cloud:** FastAPI, APIs, Streamlit, Docker, Kubernetes, Azure, AWS, Git, GitLab CI/CD, Postgres, MongoDB, Neo4j, vector databases

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)